

IS 302-2-203 : 1994

भारतीय मानक

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घरेलू और सम्बद्ध विद्युत साधित्रों की सुरक्षा

भाग 2 विशेष अपेक्षाएँ

अनुभाग 203 आन्तरिक उपयोगों के लिए विद्युत-सेल-घंटी तथा बज्जर

Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 203 Electric Call Bells and Buzzers for Indoor Use

UDC 643.62 : 654.923-83

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Price Group 3

Electrical Appliances Sectional Committee, ETD 32

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Appliances Sectional Committee had been approved by the Electrical Division Council.

This standard covers general and safety requirements of Electric Call Bells and Buzzers for indoor use. The performance requirement including general and safety requirements are covered under a separate Indian Standard IS 2268 : 1988 Specification for electric call bells and Buzzers for indoor use.

This standard is to be read in conjunction with IS 302-1 (1979). For the sake of convenience, the clauses of this standard correspond to those of IS 302-1 (1979). Instead of reproducing full text of each clause, clauses of IS 302-1 (1979) which are applicable (which means that relevant provisions of the clause apply) or not applicable and the sub-clauses or portions thereof which are not applicable are indicated as under:

- a) In case of clause, where it is applicable or not applicable, the wording used is 'This clause of IS 302-1 (1979) is applicable/not applicable'.
- b) In case of a sub-clause or part thereof 'Not applicable'.

Wherever a sub-clause of IS 302-1 (1979) is to be replaced by a new text, it has been indicated as under:

'Replacement — followed by the new text'.

Any addition to the existing provisions of a sub-clause of IS 302-1 (1979) has been indicated as under:

'Addition — followed by the text of the additional matter'.

Clauses/tables which are additional to those of IS 302-1 (1979) are numbered starting from 101 and additional sub-clauses are numbered with the main clauses number followed by 101, 102, etc, for example 7.101.

Should however, any deviation exists between IS 302-1 (1979) and this standard, the provisions of the latter shall apply.

While preparing this standard assistance was derived from IEC 335-1 (1976) Safety of household and similar electrical equipment : Part 1 General requirement, issued by International Electrotechnical Commission.

The Indian Standards which are necessary adjuncts to this standard are given in Annex AA.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 203 Electric Call Bells and Buzzers for Indoor Use

1 SCOPE

This clause of IS 302-1 (1979) is applicable except as follows:

1.1 This standard covers the general and safety requirements for electrically operated call bells and buzzers for indoor use designed for connection to supplies at voltages not exceeding 250 V ac single phase 50 c/s or dc.

2 TERMINOLOGY

This clause of IS 302-1(1979) is applicable except as follows:

2.8 to 2.11, 2.18, 2.19, 2.21 to 2.25 and 2.28 not applicable.

2.29 Replacement

Conditions of Adequate Heat Discharge — The conditions which apply when the electric call bell and buzzer is operated under normal condition of use.

2.30 to 2.33, 2.37 to 2.41 and 2.46 not applicable.

Additional sub-clauses

2.101 Call Bell

A signalling device in which a hammer is actuated so as to strike a gong, bell, pipe, strip, etc.

2.102 Buzzer

A signalling device similar to an electric bell but without a hammer or gong, and serving to produce sound by the vibration of an armature.

3 GENERAL REQUIREMENT

This clause of IS 302-1 (1979) is applicable except as follows:

3.1 Para 2 not applicable.

4 GENERAL NOTES ON TESTS

This clause of IS 302-1 (1979) is applicable except as follows:

4.3 Note 2 and **4.7 to 4.14** not applicable.

5 RATING

This clause of IS 302-1 (1979) is applicable except as follows:

5.1 Replacement

Rated Voltage — The rated voltage shall not exceed 250 V and shall preferably be 240 V.

5.2 Not applicable.

6 CLASSIFICATION

This clause of IS 302-1 (1979) is applicable.

7 MARKING

This clause of IS 302-1 (1979) is applicable except as follows:

7.1 Replacement

Each electric call bell and buzzer shall be marked with the following information.

- a) Rated voltage;
- b) Symbol for nature of supply, if applicable;
- c) Rated frequency;
- d) Rated input in watts;
- e) Maker's or vendor's name or trade-mark;
- f) Model or type reference;
- g) Symbol for Class II construction, if applicable;
- h) Symbol for degree of protection, if applicable; and
- j) Country of origin.

7.2 and 7.4 to 7.12 not applicable.

7.101 Additional Sub-clause

The electric call bells or buzzers may also be marked with the Standard Mark.

7.102 The use of the Standard Mark is governed by the provisions of Bureau of Indian Standards Act, 1986 and the Rules and Regulations made thereunder. The details of conditions under which the licence for the use of Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

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8 PROTECTION AGAINST ELECTRIC SHOCK

This clause of IS 302-1 (1979) is applicable except as follows:

8.2 to 8.9 not applicable.

9 STARTING OF MOTOR OPERATED APPLIANCES

This clause of IS 302-1 (1979) is not applicable.

10 INPUT AND CURRENT

This clause of IS 302-1 (1979) is not applicable.

11 TEMPERATURE-RISE

This clause of IS 302-1 (1979) is applicable except as follows:

11.2 to 11.6 not applicable.

11.7 Replacement

The call bell or buzzer is operated on consecutive cycles of operation until steady conditions are established under normal conditions of use.

11.8 Para 1 and 2 Replacement

During the test, the temperature-rise shall be monitored continuously and shall not exceed the values given in Table 1 and the sealing compound, if any, shall not flow out.

The measurements specified in 13.1 are made at the end of this test.

11.9 and 11.10 not applicable.

Additional Sub-clause

11.101 Call bells or buzzers with contact breakers shall satisfy this additional requirement that with the rated voltage applied to the bell or buzzer and with the contact breaker short-circuited for a period of five minutes, the temperature of the winding shall not exceed 100°C (the reference ambient temperature being 40°C).

12 OPERATION UNDER OVERLOAD CONDITIONS OF APPLIANCES WITH HEATING ELEMENTS

This clause of IS 302-1 (1979) is not applicable.

13 ELECTRICAL INSULATION AND LEAKAGE CURRENT AT OPERATING TEMPERATURE

13.1 Replacement

The electrical insulation of the appliance at operating temperature shall be adequate and the leakage current

in normal use shall not be excessive, compliance is checked by tests given in 13.2. The tests are made when the call bell or buzzer is connected to the supply.

13.4 Not applicable.

14 RADIO AND TELEVISION INTERFERENCE SUPPRESSIONS

14.1 Replacement

In case the electric call bell or buzzer contains a contact breaker, it shall include a radio interference suppression capacitor. This shall, however, not adversely affect the safety of the appliance.

15 MOISTURE RESISTANCE

This clause of IS 302-1 (1979) is applicable except as follows:

15.1 to 15.3 not applicable

16 INSULATION RESISTANCE AND ELECTRIC STRENGTH (AFTER HUMIDITY TREATMENT)

This clause of IS 302-1 (1979) is applicable except as follows.

16.1 Replacement

The insulation and electric strength of the appliances shall be adequate.

Compliance is checked by tests given in 16.2 and 16.4 which are made on the call bell or buzzer after the test in 15.4 in the humidity cabinet or in the room in which the sample was brought to the prescribed temperature after reassembly of those parts which may have been removed.

16.2 Replacement

An ac test voltage is applied as specified in items 1 to 4 of the table of 16.4. The metal foil having a size not exceeding 20 × 10 cm and being moved, if necessary, so as to test all parts of the surface.

The test voltage is 1.06 times the upper limit of the rated voltage range for dc only, for single-phase appliances and for three-phase appliances which are also suitable for single-phase supply, if the rated voltage or the upper limit of the rated voltage does not exceed 250 V.

The leakage current measured after the application of the test voltage shall not exceed the value given in 13.2.

16.3 Not applicable.

16.4 Para 1 Replacement

Immediately after the test given in 16.2, the insulation is subjected for 1 minute to a voltage of substantially

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sine wave form having a frequency of 50 or 60 Hz. The value of the test voltage and the points of application are shown in the table.

17 OVERLOAD PROTECTION

This clause of IS 302-1 (1979) is not applicable.

18 ENDURANCE (under consideration)

19 ABNORMAL OPERATION

This clause of IS 302-1 (1979) is not applicable.

20 STABILITY AND MECHANICAL HAZARDS

This clause of IS 302-1 (1979) is not applicable.

21 MECHANICAL STRENGTH

This clause of IS 302-1 (1979) is applicable except as follows:

21.2 to 21.4 not applicable.

22 CONSTRUCTION

This clause of IS 302-1 (1979) is applicable except as follows:

22.4, 22.7, 22.9 to 22.12, 22.15, 22.16, 22.18, 22.20, 22.23 to 22.27 and 22.29 to 22.32 not applicable.

23 INTERNAL WIRING

This clause of IS 302-1 (1979) is applicable except as follows:

23.4 and 23.5 not applicable.

24 COMPONENTS

This clause of IS 302-1 (1979) is applicable except as follows:

24.2 to 24.10 not applicable.

25 SUPPLY CONNECTION, AND EXTERNAL FLEXIBLE CABLES AND CORDS

This clause of IS 302-1 (1979) is applicable except as follows:

25.3 to 25.7 and 25.10 to 25.14 not applicable.

26 TERMINAL FOR EXTERNAL CONDUCTORS

This clause of IS 302-1 (1979) is applicable.

27 PROVISION FOR EARTHING

This clause of IS 302-1 (1979) is applicable except as follows:

27.3 not applicable.

28 SCREWS AND CONNECTIONS

This clause of IS 302-1 (1979) is applicable.

29 CREEPAGE DISTANCES AND CLEARANCES

This clause of IS 302-1 (1979) is applicable.

30 RESISTANCE TO HEAT, FIRE AND TRACKING

This clause of IS 302-1 (1979) is applicable.

31 RESISTANCE TO RUSTING

This clause of IS 302-1 (1979) is applicable.

32 RADIATION HAZARD

This clause of IS 302-1 (1979) is applicable.

101 TESTS

101.0 Categories of Tests

Tests are classified as type, acceptance and routine tests.

101.1 Type Tests

The tests specified in Table 101 shall institute the type tests and shall be carried out on two samples of call bell or buzzer of the same type and rating selected, preferably at random, from a regular production lot. Before commencement of the tests, the call bell or buzzer shall be visually examined and inspected for obvious visual defects in respect of components, parts and their assembly, construction, stability, markings, provision of suitable terminals for supply connections, earthing, and the effectiveness of screws and connections. The external surface finish shall be even and free from finishing defects.

101.1.1 Criteria of Acceptance

All samples shall successfully pass all the type tests for proving conformity with the requirements of the standard. If any of the samples fails in any of the type tests, the testing authority, at its discretion, may call for fresh samples not exceeding twice the original number and subject them again to all tests or to the test(s) in which failure(s) occurred. No failure should be permitted in the repeat test(s).

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101.2 Acceptance Tests

The following shall constitute the acceptance tests:

<i>Test</i>	<i>Clause Reference</i>
i) Protection against electric shock	8
ii) Temperature rise	11
iii) Insulation resistance and leakage current at operating temperature	13
iv) Moisture resistance	15
v) Insulation resistance and electric strength (after humidity treatment)	16
vi) Provision for earthing	27

101.2.1 A recommended sampling procedure for acceptance tests is given in Appendix B of IS 302-1 (1979).

101.2(1).2(3) Routine Tests

The following tests shall constitute the routine tests:

<i>Test</i>	<i>Clause Reference</i>
a) Protection against electric shock	8
b) High voltage	13.3.2 of IS 302-1 (1979)
c) Provision for earthing	27

Table 101 Schedule of Type Tests

(Clause 101.1)

Sl No.	Test	Clause Reference
i)	Protection against electric shock	8
ii)	Temperature-rise	11
iii)	Electrical insulation and leakage current at operating temperature	13
iv)	Radio and television interference suppression	14
v)	Moisture resistance	15
vi)	Insulation resistance and electric strength (after humidity treatment)	16
vii)	Endurance (under consideration)	18
viii)	Mechanical strength	21
ix)	Construction	22
x)	Internal wiring	23
xi)	Component	24
xii)	Supply connection and external flexible cables and cords	25
xiii)	Terminal for external conductor	26
xiv)	Earthing	27
xv)	Screws and connections	28
xvi)	Creepage distances and clearance	29
xvii)	Resistance to heat, fire and tracking	30
xviii)	Resistance to rusting	31
xix)	Radiation hazards	32

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ANNEX A

TABLES OF TYPE TESTS

Appendix A of IS 302-1 (1979) is not applicable.

ANNEX B

SAMPLING PROCEDURE FOR ACCEPTANCE TEST

Appendix B of IS 302-1 (1979) is applicable.

ANNEX C

ELECTRONIC CIRCUITS

Appendix C of IS 302-1 (1979) is applicable, if electronic circuits are used.

ANNEX D

MEASUREMENT OF TEMPERATURE WITH THERMOMETER

Appendix D of IS 302-1 (1979) is applicable.

ANNEX E

ALTERNATIVE TESTS FOR PROTECTED MOTOR UNITS

Appendix E of IS 302-1 (1979) is not applicable.

ANNEX F

IMPACT TEST APPARATUS

Appendix F of IS 302-1 (1979) is applicable.

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ANNEX G

THERMAL CENTRAL AND OVERLOAD RELEASES

Appendix G of IS 302-1 (1979) is not applicable.

ANNEX H

MEASUREMENT OF CREEPAGE DISTANCES AND CLEARANCES

Appendix H of IS 302-1 (1979) is applicable.

ANNEX J

TEST FOR FIRE-RESISTING PROPERTIES

Appendix J of IS 302-1 (1979) is applicable.

ANNEX K

BNF JET TEST FOR DETERMINATION OF THICKNESS OF COPPER AND NICKEL PLATING

Appendix K of IS 302-1 (1979) is not applicable.

ANNEX L

APPROXIMATE MEASUREMENT OF THICKNESS OF CHROMIUM ON NICKEL, STEEL AND COPPER

Appendix L of IS 302-1 (1979) is not applicable.

ANNEX AA

<i>IS No.</i>	<i>Title</i>
302-1(1979)	Safety of household and similar electrical appliances : Part 1 General requirements (<i>fifth revision</i>)

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